

**{{ QUESTION\_TITLE: Line with Slope 2/3 Passing Through Point (4, -1) }}**

## ORIGINAL EXAM QUESTION / ATTEMPT

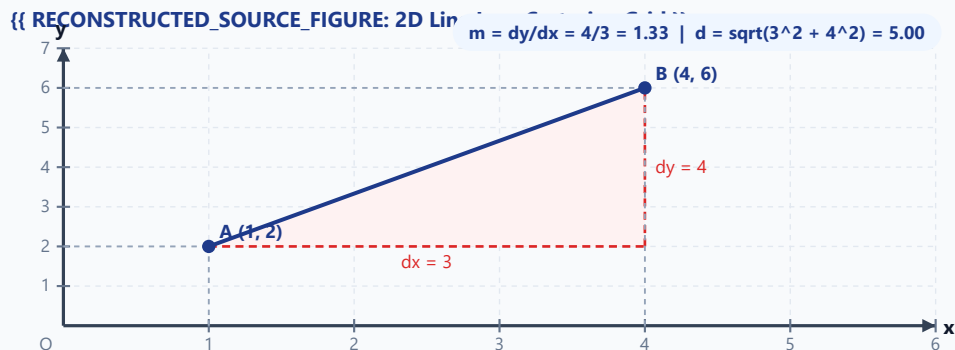
**{{ QUESTION\_STEM }}**:

A straight line  $L$  passes through point  $A(4, -1)$  with slope  $m = 2/3$ .

Which of the following points also lies on line  $L$ , and what is the  $y$ -intercept of  $L$ ?

Options:

- (a) Point  $(-2, -5)$  and y-intercept  $= -11/3$       (b) Point  $(1, -3)$  and y-intercept  $= -11/3$   
(c) Point  $(7, 1)$  and y-intercept  $= 11/3$       (d) Point  $(2, -2)$  and y-intercept  $= -8/3$



**{{ H0\_ATTEMPT\_WORKSPACE: Work Here \* Try Before Looking at Hints }}**

**Your Final Choice:** [       ]    **Reason:**

**H1-H3 HINT LADDER (READ ONE AT A TIME ONLY IF STUCK)**

\* {{ H1\_NOTICE }}: The slope  $m = (y - y_1) / (x - x_1)$  is identical for ALL points on the line.

\* {{ H2\_MODEL }}: Use point-slope form:  $(y - (-1)) = 2/3 (x - 4) \Rightarrow 3(y + 1) = 2(x - 4)$ .

Rearrange into standard linear equation form:  $2x - 3y - 11 = 0$ .

\* {{ H3\_START }}: Substitute candidate points into  $2x - 3y - 11 = 0$  to see which gives 0.

For y-intercept, evaluate at  $x = 0$ :  $2(0) - 3y - 11 = 0 \Rightarrow y = -11/3$ .

## SEMANTIC REASONING ROUTE

**Step 1 [INTERPRET]:** Identify given geometric parameters: point  $(4, -1)$ , slope  $m = 2/3$ .

**Step 2 [MODEL & BIND]:** Convert to algebraic form using point-slope invariant:  $2x - 3y - 11 = 0$ .

**Step 3 [DISCRIMINATE]:** Test options:  $(-2, -5) \Rightarrow 2(-2) - 3(-5) - 11 = -4 + 15 - 11 = 0$ . [VALID]

**Step 4 [VERIFY INTERCEPT]:** Set  $x = 0 \Rightarrow y = -11/3$ . Select matching option (a).

**VERIFIED SOLUTION & CHECK**

**{{ SOLUTION\_FINAL\_ANSWER }}**: Correct Option is (a).

### Execution Walkthrough:

1. Standard Equation:  $2x - 3y - 11 = 0$ .
2. Option (a) Point Check: Substitute  $(-2, -5)$ :  
 $LHS = 2(-2) - 3(-5) - 11 = -4 + 15 - 11 = 0 = RHS$ . (Satisfies L)
3. Intercept Determination:  
L crosses y-axis where  $x = 0 \Rightarrow 2(0) - 3y - 11 = 0 \Rightarrow -3y = 11 \Rightarrow y = -11/3$ .

**{{ VERIFICATION\_CHECK }}:**

Compute slope between given point A(4, -1) and discovered point B(-2, -5):

$m_{\text{calc}} = (-1 - (-5)) / (4 - (-2)) = 4 / 6 = 2/3$ . Exactly matches problem statement.

**Permanent Invariant:** Points lie on a line if and only if they satisfy its equation.